

Simulated ocular biometry effects on add-power demand in multifocal intraocular lenses



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Purpose: To use ray-tracing simulations to assess the dependency of add-power demand on ocular biometry.

Setting: David J. Apple Laboratory for Ocular Pathology, Department of Ophthalmology, University of Heidelberg, Heidelberg, Germany.

Design: Experimental study.

Methods: Optical biometry of 34 patients scheduled for routine cataract surgery was collected and the following parameters used to build a personalized eye model: keratometry (K), anterior chamber depth (ACD), and axial length (AL). Ray-tracing software was applied for pseudophakic-eye simulations. Each model featured a biconvex lens with radii of curvature adjusted for best distance vision. Trial spectacles with -4 diopters (D), -3 D, -2 D, and -1.5 D were placed in front of a virtual eye. The intraocular lens (IOL)-power change required to induce a desirable near/intermediate effect at the spectacle plane was calculated for the far-point condition.

Results: Mean values for K, ACD, and AL were 7.79 ± 0.24 mm, 3.22 ± 0.60 mm, and 23.77 ± 1.34 mm, respectively. The lowest and highest add power (IOL plane) varied by 0.39 D, 0.52 D, 0.76 D, and 0.98 D for -1.5 D, -2 D, -3 D, and -4 D trial spectacles, respectively. High correlation was found between the observed add-power change and the ACD, indicating a linear increase of the add-power demand for larger ACD values. The AL and the K demonstrated poor or no correlation with the required IOL-power adjustment.

Conclusions: The ACD, which determines the effective lens position in our model, has greater impact on add-power demand than AL or K. The reported change may contribute to interparticipant variability in defocus curves of multifocal patients.

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Multifocal intraocular lenses (IOLs) can provide spectacle independence for patients undergoing cataract or refractive lens exchange surgery. Early models, such as bifocal lenses, were designed to achieve functional visual acuity (VA) at near and distance ranges but lacked intermediate clarity.¹ Trifocal lenses soon followed, adding intermediate focus to enhance vision across all distances.² More recently, extended depth-of-focus lenses have been developed, providing smoother transitions between different focal points and reducing visual disturbances such as halo or glare.³

Currently available multifocal IOLs use different optical designs and addition powers to generate multifocality, which may vary between models and manufacturers.⁴ However, although near addition powers are always fixed, the visual outcomes may differ between patients who

underwent implantation with the exact same lens model. Such differences become more clinically evident when the postoperative visual outcomes are assessed through defocus curves: clinical studies have shown high interparticipant variability in the positions of the secondary VA peaks for the same IOL.^{5–7}

Although the exact mechanism remains elusive, one possible source of such variability may stem from the differences in ocular biometry. In particular, keratometry (K), axial length (AL), and anterior chamber depth (ACD) or effective lens position (ELP) have been implicated to play a role.^{5–7}

Thus, in this experimental study, we performed ray-tracing simulations to predict the add-power shift of the secondary VA peaks arising from ocular biometry differences in pseudophakic eyes.

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Table 1. Patient data

Parameter	Age (y)	SE (D)	CDVA (logMAR)
Mean	72	−0.19	0.25
SD	7	2.55	0.26
Min	57	−7.75	−0.04
Max	80	3.5	1.1

SE = spherical equivalent

METHODS

Participants

This laboratory study collected biometry measurement reports (IOL Master 500, Carl Zeiss Meditec AG) of 34 patients scheduled for routine cataract surgery. Patient age, spherical equivalent, and corrected distance visual acuity (CDVA) are given in Table 1. Assessed parameters included corneal K (in millimeters), ACD (in millimeters), and AL (in millimeters). Only measurements with acceptable quality were exported for analysis.

Ray-Tracing Simulation

A virtual eye model was created using OpticStudio (Radiant Zemax LLC) based on the anatomical model described by Liou and Brennan, which was modified to account for patients' individual radii of curvature, ACD, and AL.⁸ First, for each eye, an aphakic model was built. Then, a biconvex lens was placed with radii of curvature adjusted for best distance vision. The position of the IOL in the model eye was determined by the ACD, that is, the distance from the cornea based on biometry data. Subsequently, trial spectacles with −4.0 diopters (D), −3.0 D, −2.0 D, and −1.5 D were placed with a vertex distance of 12 mm. The change in IOL power required to induce a desirable near/intermediate effect at the spectacle plane was calculated for the far-point condition of each eye.

Statistical Analysis

Statistical analyses were performed using packages statistics and machine learning toolbox in MATLAB R2024a (Mathworks, Inc.). Correlation analysis was performed using the Pearson correlation coefficient (denoted r), and linear regression analysis was used to model the relationships between add-power at the IOL plane and K, ACD, and AL, with the coefficient of determination R^2 provided.

RESULTS

Ocular Biometry

In this population of 34 patients with cataract, the mean $\pm$ SD values for K, ACD, and AL were 7.79 ± 0.24 mm, 3.22 ± 0.60 mm, and 23.77 ± 1.34 mm, respectively.

Add Powers

For trial spectacles −4.0 D, −3.0 D, −2.0 D, and −1.5 D, the maximum differences between the lowest and highest addition powers at the IOL plane were 0.98 D, 0.76 D, 0.52 D, and 0.39 D, respectively.

Correlations

For add power of +1.5 D, a high positive correlation ($r = 0.98$, $P < .001$) was found between the observed add-power change and the ACD, but the AL ($r = 0.31$, $P = .07$) and the K ($r = -0.01$, $P = .97$) demonstrated a weak positive or no correlation, respectively (Figure 1). The same trend was observed for add power of +2.0 D, where a high correlation was found between the observed add-power change and the ACD, and no significant correlation with neither AL nor K (Figure 2). Similarly, for add powers of +3.0 D and +4.0 D, we found significant association between the required add-power demand and the ACD, but no significant correlations with AL and K (Figures 3 and 4).

DISCUSSION

The efficacy of multifocal IOLs in providing postoperative spectacle independence has led to their increased popularity and demand worldwide. Careful patient selection and counselling are essential for optimal visual outcomes and patient satisfaction. Knowledge of patients' occupation, hobbies, and primary needs affects the type of multifocal lens model implanted to select the appropriate addition power(s) tailored to their demands. The interparticipant variability in secondary VA peaks reported in recent literature warrants particular attention because such variations in the spectacle plane of the add power may lead to inadequate reading distance and patient dissatisfaction.

IOL calculation for cataract or refractive lens exchange surgery relies on biometry data such as AL, ACD, and K.

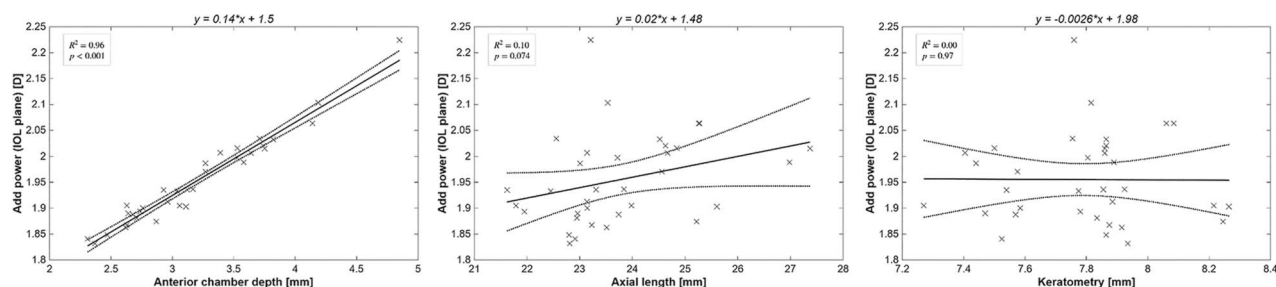


Figure 1. Linear association between the required add-power shift for +1.5 D near addition and anterior chamber depth, axial length, and corneal keratometry demonstrates statistically significant relationship only for anterior chamber depth. Dotted lines = 95% CI; R^2 = coefficient of determination; P = significance level.

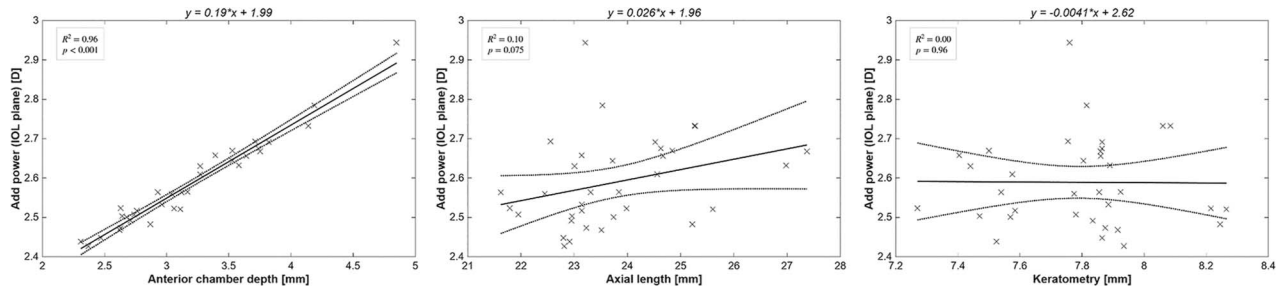


Figure 2. Linear association between the required add-power shift for +2.0 D near addition and anterior chamber depth, axial length, and corneal keratometry demonstrates statistically significant correlations only for anterior chamber depth. Dotted lines = 95% CI; R^2 = coefficient of determination; P = significance level.

Depending on the refractive status of the eye, there may be differences in their biometric parameters: myopic eyes tend to have longer AL, deeper ACD, and higher K, whereas hyperopic eyes are typically associated with shorter AL, shallower ACD, and lower K. Furthermore, myopic eyes often have thinner lens thickness (LT) than hyperopic eyes which affects the calculation of the ELP.^{9,10}

Predicting the ELP is a complex task due to its reliance on preoperative biometric factors and intraoperative conditions. As such, several fourth generation and artificial intelligence-based formulas or ray-tracing models have been introduced to allow more precise ELP predictions and enhance the postoperative refractive outcomes.^{11,12}

Biometric parameters and IOL calculation become even more relevant when a multifocal IOL is planned. The add powers of a given multifocal lens dictate the near or intermediate distances at which the patients can enjoy spectacle independence and they range from +2.5 to +4.0 D.⁶ Naturally, careful prediction of the correct IOL base power can ensure uncorrected vision at the promised distances.

In a theoretical study, Savini et al. reported that eyes with longer AL and steeper K have the longest near focal distance after implantation with multifocal IOLs compared with those with shorter AL and flatter K who have the shortest near focal distance.⁷ Such differences in near focal distance despite the same multifocal lens model implanted have been shown to occur as a function of ELP, where AL and K play an indirect role.^{7,13} A longer ELP would cause a decrease in nominal power of the IOL at its spectacle plane, which in

turn would also decrease its near addition power. Their theoretical analysis was corroborated by clinical evidence, where Petermeier et al. assessed defocus curves of 32 patients implanted with Restor multifocal IOL with +4.0 D near addition and observed mean reading distances of 29.5 cm, 32.8 cm, and 34.6 cm for hyperopic, emmetropic, and myopia patients, respectively.⁵ The authors found statistically significant correlations between the reading distance and AL and ACD, underlining the importance of preoperative evaluation of the biometric components.

This study is not without limitations. First, given the small sample size and the lack of short and long eyes, future studies with larger sample size and wider range of AL are necessary to corroborate our findings. Second, our analysis did not take into account the effects of pupil size, LT, or anterior segment length, that is, the sum of ACD and LT, which may also contribute to predicting the required dioptric power for near and intermediate vision; further research is warranted to investigate their impact. Furthermore, we used the preoperative ACD as the postoperative ELP because the purpose of this study was not to test the accuracy of any IOL power calculation or predict the postoperative lens position but rather to determine the effect of ACD on the add power of simulated lenses. Although this remains a limitation, the range of ACD measured in our study is comparable and within the range of the postoperative ACD values found in the literature.^{14–16} Moreover, given the high correlation and strong linear relationship observed in our results, any expected

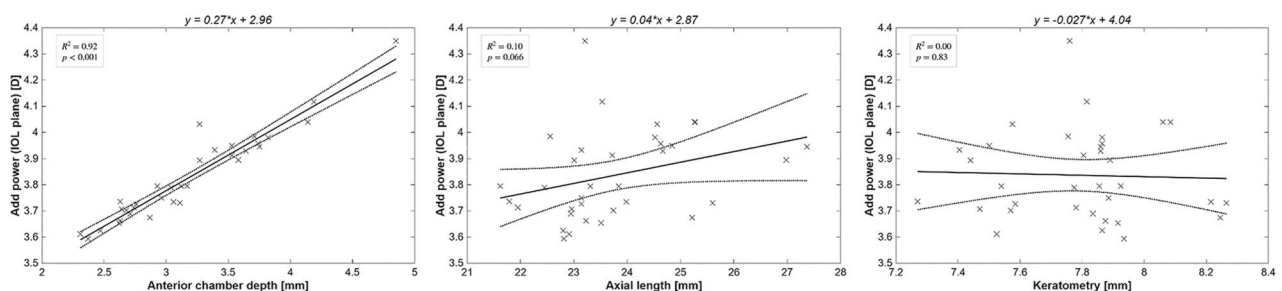


Figure 3. Linear association between the required add-power shift for +3.0 D near addition and anterior chamber depth, axial length, and corneal keratometry demonstrates statistically significant correlations only for anterior chamber depth. Dotted lines = 95% CI; R^2 = coefficient of determination; P = significance level.

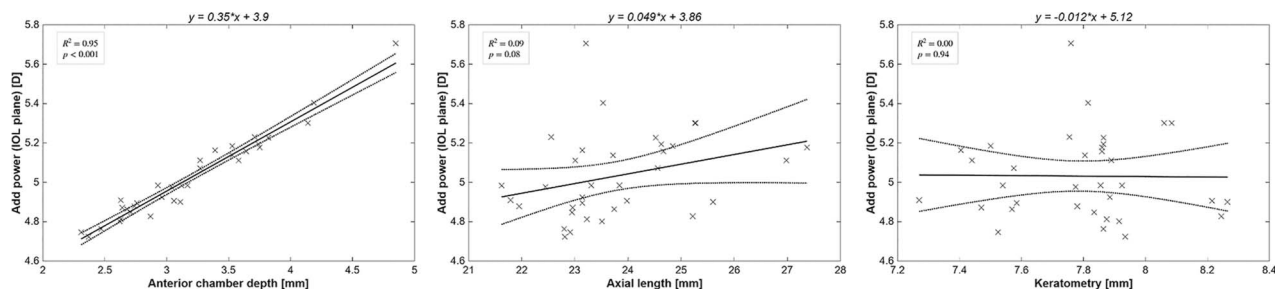


Figure 4. Linear association between the required add-power shift for +4.0 D near addition and anterior chamber depth, axial length, and corneal keratometry demonstrates statistically significant correlations only for anterior chamber depth. Dotted lines = 95% CI; R^2 = coefficient of determination; P = significance level.

postoperative ACD value can be entered into the regression formula to predict its effect on near vision outcomes after multifocal IOL implantation.

In this study, we used ray-tracing simulations to investigate correlations between ocular-biometric parameters and the required add-power shift to attain the optimal VA for secondary peaks and found statistically significant correlations between ACD and all studied add-powers. In accordance with previous findings, AL and K have not been found to contribute significantly to add-power shift in our model.^{17–20}

Thus, we provide compelling statistical evidence that patients with shallower ACDs (eg, hyperopes) may have their near focus closer than preoperatively planned and a higher add-power lens may be required for eyes with a deeper anterior chamber (eg, myopes). Our results presented herein therefore further corroborate and echo previous recommendations that refractive statuses and ACD, or ELP, should be taken into consideration when planning a multifocal IOL implantation.

WHAT WAS KNOWN

- Multifocal IOLs use different optical designs and addition powers to generate multifocality, which may vary between models and manufacturers.
- Interestingly, although near-addition powers are always fixed, the secondary VA peak may differ between patients who underwent implantation with the exact same lens model.
- Such high interparticipant variability in the positions of secondary VA peaks among eyes implanted with the same IOL may arise from differences in optical biometry.

WHAT THIS PAPER ADDS

- This experimental study used ray-tracing simulations to predict the add-power shift of the secondary VA peaks arising from ocular biometry differences in pseudophakic eyes and found statistically significant correlations between anterior chamber depth (ACD) and add-powers, while keratometry and axial length were not significantly correlated.
- We provide compelling statistical evidence that patients with shallower ACDs (eg, hyperopes) may have their near focus closer than preoperatively planned and a higher add-power lens may be required for eyes with a deeper anterior chamber (eg, myopes).

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